Machine Learning for Business

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Key Points of the Course

Objective

- Understand the core principles of Machine Learning
- Connect theory to real business decision-making

Learning Outcomes

- Select appropriate models depending on the problem
- Interpret results and evaluate performance
- Understand limitations and trade-offs

Assessment & Practice

- Applied exercises throughout the course
- Mid-term exam (March 10th)
- Final group business project

Course Roadmap — Machine Learning for Business

Module 0 — Big Picture

- What is Machine Learning?
- Supervised vs Unsupervised Learning
- From Data to Decision-Making

What is Machine Learning?

Definition: Machine Learning (ML) enables algorithms to learn patterns and tasks directly from data without being explicitly programmed.

Key Points:

- ML lies at the intersection of **Mathematics** and **Computer Science**.
- A **Machine Learning Model** automates repetitive tasks.
- An **AI Agent** goes further: it takes decisions and actions based on triggers and observations.
- Two main strategies to teach a Machine Learning Model:
  - *Supervised Learning*: The model is guided using labeled data points (x_i, y_i).
  - *Unsupervised Learning*: The model discovers structure or patterns in unlabeled data (x_i).

Supervised Learning

Setup: We have input-output pairs:

$$(x_i, y_i), \quad i = 1, \ldots, n$$

Goal: Learn a function $f(x)$ that predicts y accurately.

Two main types:

- *Regression:* Predict a continuous outcome (e.g., house prices, sales forecast)
- *Classification:* Predict a categorical outcome (e.g., loan default, churn yes/no)

Business Example: A bank wants to predict whether a new applicant will default (classification) or estimate the expected loan amount (regression).

Unsupervised Learning

Goal: Discover structure in unlabeled data.

We observe only features:

$$x_i, \quad i = 1, \dots, n$$

No target y is available.

Common Tasks:

- *Clustering*: Group similar data points (e.g., customer segmentation)
- *Dimensionality Reduction*: Summarize data with fewer variables (e.g., PCA for marketing analytics)

Business Example: A retail company wants to identify customer segments based on purchase behavior to design targeted marketing campaigns.

Data-Driven Approach

Idea: Use data to drive business decisions instead of intuition alone.

Steps:

1. Collect and clean relevant data
2. Explore and visualize patterns
3. Build predictive or descriptive models
4. Make decisions and take action
5. Monitor outcomes and iterate

Business Example: A bank wants to optimize credit offers:

- Analyze past customer data
- Predict likelihood of accepting a new offer
- Adjust campaigns dynamically

Module 1 — Probability Foundations

- Random Variables (Discrete and Continuous)
- Probability Distributions
- Expectation and Variance
- Normal Distribution
- Law of Large Numbers
- Central Limit Theorem
- Confidence Intervals
- Bayes' Theorem

Why Probability in Business?

Business Example: Loan Default
A bank evaluates loan applications.

- 8% of all customers default
- 25% of customers with low credit score default

Question:

What is the probability that a new applicant will default?

Key Idea: Business decisions are made under uncertainty.

Why We Need Probability in Machine Learning

In machine learning, we try to predict an output based on input variables such as education, experience, age, or other business characteristics.

However, in real life the same inputs do not always lead to the same output.

- Two people with the same education and experience may earn different wages.
- Two customers with identical profiles may behave differently.
- Two products with the same features may still sell at different prices.

This variation comes from many sources:

- unobserved factors,
- randomness in human behaviour,
- measurement errors,
- missing information,
- natural economic and social variability.

Why Probability Is Necessary

Because outcomes vary even when inputs are identical, machine learning models must treat outputs as **random variables** rather than fixed quantities.

Your probability tools explain how to describe such randomness:

- A random variable represents all possible outcomes.
- Probability tells us how frequently each outcome occurs.
- Expected value describes the long-run average.
- Variance measures uncertainty and risk.
- The Law of Large Numbers explains why estimates improve with more data.
- The Central Limit Theorem explains why averages behave predictably.

Probability and Model Evaluation

Probability theory is essential for understanding and evaluating machine learning models:

- It explains why models cannot predict perfectly.
- It allows us to estimate and interpret prediction errors.
- It provides the foundation for loss functions used in training.
- It helps us understand how models generalize to unseen data.
- It quantifies uncertainty in business predictions.

Machine learning does not predict exact outcomes. It predicts patterns in data that are inherently variable. Probability allow us to evaluate these predictions.

Set of all outcomes Ω

We imagine an Excel file containing rows such as: experience, education, wage, gender, etc.

If more people are observed, the file grows: each new person adds a new row.

Ω **is defined as the collection of all possible rows** that could ever appear in this Excel file.

In **general:** Ω is the collection of all possible rows.

Each row in our dataset is one observed element of Ω. Future data are simply additional elements drawn from the same set.

Probability measure

Probability describes how often different types of rows appear.

A **event** A is a subset of Ω: collection of rows with a certain property: certain combination of values of columns.
Examples of probability events:

- Wage smaller than 10.
- Education greater than 8.
- Female $= 1$ (the worker is female).
- "Female $= 1$ AND education > 8 AND wage < 10."

Probability $P(A)$ **of an event** A: how often we see rows from A

Probability $P(A)$ of an event A: a more precise defintion

Choose N rows from Ω, N large

$$P(A) = \frac{\text{number of rows among the chosen } N \text{ rows which are in } A}{N}$$

We want $P(A)$ to be roughly the same as N grows: limit exists

$$P(A) = \lim_{N \to \infty} \frac{\text{number of rows among the chosen } N \text{ rows which are in } A}{N}$$

The elements of Ω should be **independent samples**: the frequencies

$$\frac{\text{number of rows among the chosen } N \text{ rows which are in } A}{N}$$

should be approximately for the same N for any choice of N rows.

If rows correspond to time (time-series), not always true.

Probability: difficulties

The precise definition is difficult: Kolmogorov 1930's, advanced math.

Properties:

- $P(A) \geq 0$ for all events A,
- $P(\Omega) = 1$,
- If A and B cannot occur at the same time, then $P(A \cup B) = P(A) + P(B)$.

Probability is not always defined The frequency

$$\frac{\text{number of rows among the chosen } N \text{ rows which are in } A}{N}$$

need not stabilize as N grows.

We can never know based on data if probability is defined, we assume it.

We can make some experiments to see if this assumption is not invalidated.

Random variable

A **random variable** assigns a numerical value to every row in Ω.

Formally:

$$X : \Omega \to \mathbb{R}$$

Interpretation:

- Each column in the Excel file is a random variable.
- The random variable takes a row and outputs the value in that column.

Examples:

- Experience: maps each row to that person's experience.
- Education: maps each row to that person's education.
- Wage: maps each row to that person's wage.
- Gender: maps each row to 0 (male) or 1 (female).

Discrete and Continuous Random Variables

A random variable can have two types of possible values:

Discrete random variables

- Take values like 0, 1, 2, 3, ...
- There are no values in between.
- Example: gender ($0 =$ male, $1 =$ female).

Continuous random variables

- Can take any value with decimals in between.
- Example: wage (10.2, 10.21, 10.212, and so on).

The value of a random variable for a given row depends on the values in that row's columns.

Probability Distributions

A probability distribution describes how the values of a random variable are spread across all possible rows in Ω.

Discrete case $p(a)$: event $X = a$ – all rows to which X assigns a.

$$p(a) = \mathbf{P}(X = a)$$

$p(a)$ is relative frequency of rows for which X is a.
Continuous case $F(a)$: Events:$X \leq a$, $X \geq b$.

$$F(a) = \mathbf{P}(X \leq a)$$

$F(a)$ is relative frequency of rows for which $X \leq a$.

$$\mathbf{P}(a \leq X \leq b) = \begin{cases} \sum_{s=a}^{b} p(s) & \text{discrete} \\ F(b) - F(a) & \text{continuous} \end{cases}$$

Probability Density Function

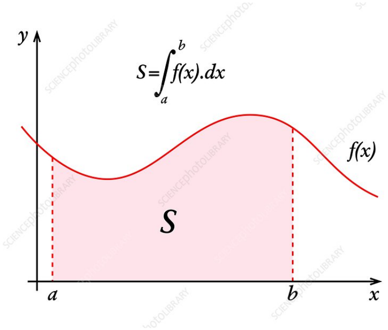
Definition: A continuous random variable has a density $f(x)$ such that:

- $f(x) \geq 0$
- $\int_{-\infty}^{\infty} f(x)\,dx = 1$
- f is the derivative of the probability distribution $\frac{dF(a)}{da} = f(a)$.

Probability of an interval:

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f(x)\,dx$$

f - density function. Integral: area under curve.



Example of density

Gaussian distribution: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}$,

uniform distribution on $[0,1]$: $f(x) = \begin{cases} 1 & x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$.

For continuous variables: $\mathbf{P}(X = a) = 0$

Example Discrete Random Variables

Business Example: Online Shop

Let

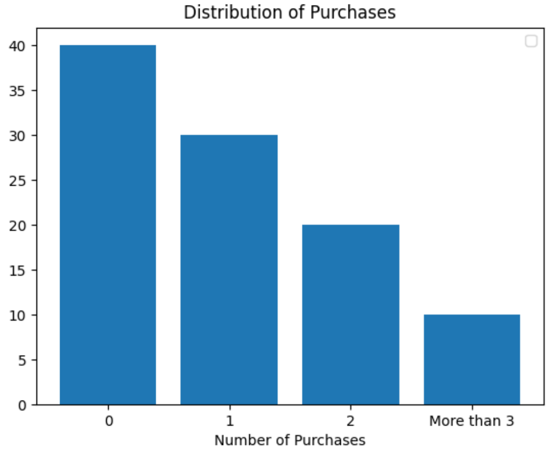
$$X = \text{Number of purchases per customer per month}$$

Possible values:

$$X \in \{0, 1, 2, 3, \dots\}$$

Example: Probability Distribution of Purchases

Histogram for Discrete Distribution:



Discrete Probability Distribution

Business Example: Purchase Behavior

Purchases	Relative Frequency
0	0.4
1	0.3
2	0.2
3+	0.1

Example: Continuous Random Variables

Business Example: Delivery Time
Let

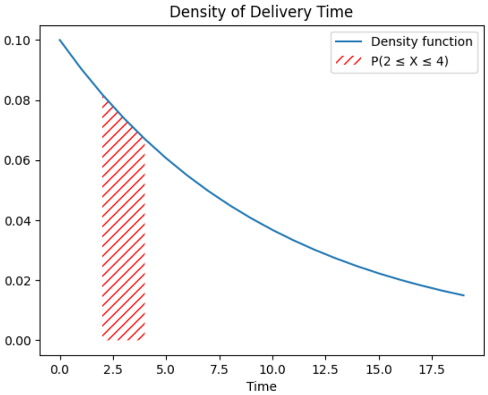
$$X = \text{Delivery time (in days)}$$

Possible values:

$$2.13,\ 3.47,\ 4.002,\ \dots$$

Assumption: The density function of delivery time follows an exponential distribution of parameter $\lambda = 0.10$ (i.e. $f(x) = \lambda\, exp(-\lambda x)$ if $x \geq 0$, else 0).

Visualization of Probability:



From samples to distribution

Value of a random variable $\mathbf{X}$ are **samples**.

Intuitively, if $x_1, \ldots, x_t$ samples of $\mathbf{X}$, (t row index), the the distribution can be computed:

$$F(a) = \hat{F}_N(a), \quad \text{for any choice } t_1 < t_2 < \ldots < t_N,\ N \text{ large}$$

empirical frequency

$$\hat{F}_N(a) = \frac{1}{N}(\text{ number of times } x_t \leq a,\ t = t_1 < \ldots < t_N)$$

Problem what if $x_1 < x_2 < \cdots < x_n$?

$x_1, \ldots, x_t$ should be **independently identically sampled (i.i.d) from** F :

Independently sampled data

$x_1, \ldots, x_n$ **i.i.d** of F, if for any choice (independent of data) of non-overlapping data

- $z_1, \ldots, z_N$,
- $y_1, \ldots, y_N$

from $x_1, \ldots, x_n$ such that N is large:

- **Identical probabilities:**

  $$\frac{1}{N}(\text{ number of times } z_i < a,\ i = 1, 2, \ldots, N) =$$
  $$\frac{1}{N}(\text{ number of times } y_i < a,\ i = 1, 2, \ldots, N) = F(a)$$

  relative frequences of events in $y_1, \ldots, y_N$, $z_1, \ldots, z_N$ is the same.

- **Independence**

$$\frac{1}{N}(\text{ number of times } z_i < a,\ y_i < b,\ i = 1, 2, \ldots, N) \approx F(a)F(b)$$

the values of y_i and z_i change independtly from each other.

Independently sampled data

if $z_1, \ldots, z_N$ are past observations, and $y_1, \ldots, y_N$ are new observations, then frequencies in the past tell us about frequencies of the future.

Independence $\implies$ frequencies of different subsets of data are the same and do not influence each other

Independent samples, independent variables

Two random variable X and Y are independent,

$$\begin{aligned} \mathbf{P}(X = a, Y = b) &= \mathbf{P}(X = a)\mathbf{P}(Y = b), \quad \text{discrete variable} \\ \mathbf{P}(X \leq a, Y \leq b) &= \mathbf{P}(X \leq a)\mathbf{P}(Y \leq b), \quad \text{continuous variable} \end{aligned}$$

The sample $x_1, x_2, \ldots, n$ is i.i.d. sample of F, if there exists sample space $\widetilde{\Omega}$ and random variables $X_1, X_2, \ldots$ on $\widetilde{\Omega}$, such that

- x_t is a sample of some random variables X_t corresponding to the same element (row) of $\widetilde{\Omega}$,
- X_t are **idenpendently identically distributed(i.i.d)**: have the same probability distribution F and they are independent (one variable contains no information on others):

Law of large number: if x_t, $t = 1, 2$, i.i.d. sample of F

$$F(a) \approx \frac{1}{N}(\text{ number of times } x_t \leq a,\ t = t_1 < \ldots < t_N)$$

Expected value of a random variable

Expected value of $\mathbf{E}[X]$:

- X takes discrete values:

$$\mathbf{E}[X] = \sum_{i=1}^{\infty} i\mathbf{P}(X = i)$$

- X has continuous values:

$$\mathbf{E}[X] = \int_{-\infty}^{+\infty} xf(x)dx$$

$$m_X = \mathbf{E}[X] \text{ – mean of } X.$$

$(X - m_X)^2$ is a random variable:

$$\sigma_X = \sqrt{\mathbf{E}[(X - m_X)^2]} \text{ standard deviation,}$$
$$\sigma_X^2 \text{ – variance of } X.$$

Example:Gaussian distribution: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}$, mean m,variance σ^2.

Expected value of a random variable

Expectation, variance depend on the distribution F of X.

x_t independently, randomly sampled from F:

$$m_X = \mathbf{E}[X] \approx \bar{x} = \frac{1}{N} \sum_{t=1}^{N} x_t$$

$$\sigma_X^2 = \mathbf{E}[(X - m_X)^2] \approx \frac{1}{N-1} \sum_{t=1}^{N} (x_t - \bar{x})^2$$

Interpretation of standard deviation:

$$\mathbf{P}(|X - m_X| > a) \leq \frac{\sigma_X^2}{a^2}$$

The smaller the standard deviation is, the more the points x_t are concentrated around the mean.

Covariance, correlation and independence

Covariance:

The covariance of two random variables X and Y is:

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

Measures how much X and Y vary together.

Correlation:

The correlation coefficient is:

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Normalized to $[-1, 1]$:

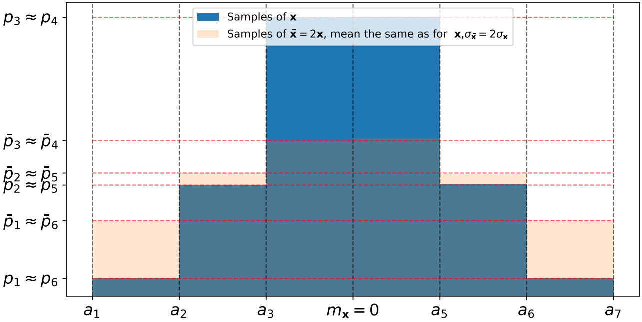
- $\rho = 1 \rightarrow$ perfect positive correlation
- $\rho = 0 \rightarrow$ no linear relationship
- $\rho = -1 \rightarrow$ perfect negative correlation

Independence and Covariance:

Mean, variance from data, visualization

np.mean(X), np.std(X), np.var(X).

Historgrams are concentrated around the mean, they get flatter with the increase of standard deviation (more points far away from the mean).



Example: Expectation of Discrete Probability Distribution

Assumption: The density function of delivery time follows an exponential distribution of parameter $\lambda = 0.10$ (i.e. $f(x) = \lambda\, exp(-\lambda x)$ if $x \geq 0$, else 0).
Business Example: Average Purchases

$$\mathbb{E}[X] = \sum_{x} x \cdot \mathbf{P}(X = x)$$

$$= 0(0.4) + 1(0.3) + 2(0.2) + 3(0.1) = 0.7$$

Interpretation:

- The purchase average is nearly 1
- It is possible to estimate the expected revenue

Example: Expectation of Probability Density Function

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x\, f(x)\, dx = \frac{1}{\lambda} = 10$$

Business Interpretation:

- The average of delivery time is 10 days
- It is possible to estimate the delivery delay between two customers.

Variance and Risk

Business Question:
Two segments have the same average revenue. Which one is riskier?

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Variance measures variability and business risk.

Normal Distribution

Definition:
A random variable X follows a normal distribution if its density is:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Notation:

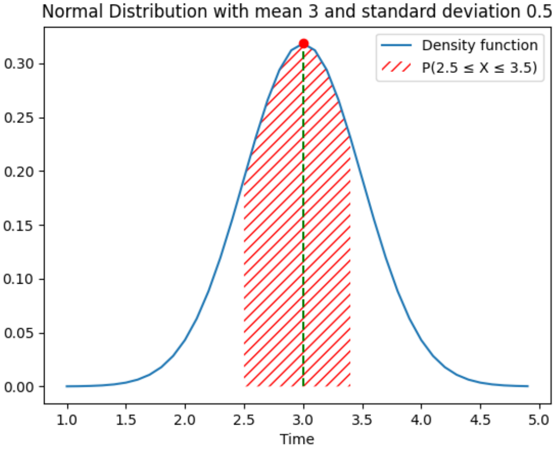
$$X \sim \mathcal{N}(\mu, \sigma^2)$$

Properties:

- $\mathbb{E}(X) = \mu$: mean
- $\mathrm{Var}(X) = \sigma^2$: variance

Normal Distribution Density

Illustration:



Example: Normal Delivery Time

Assume delivery time follows:

$$X \sim \mathcal{N}(3, 0.5^2)$$

- Mean $=$ 3 days
- Standard deviation $=$ 0.5 days

We can compute:

$$\mathbf{P}(2.5 \leq X \leq 3.5)$$

Interpretation of Parameters

Mean μ:

- Determines the center
- Shifts the distribution

Variance σ^2:

- Controls spread
- Larger variance $\Rightarrow$ flatter curve
- Smaller variance $\Rightarrow$ more concentrated

The normal distribution is symmetric around μ.

The 68–95–99.7 Rule

Empirical Rule for the Normal Distribution
If

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

then approximately:

$$\mathbf{P}(|X - \mu| \leq \sigma) \approx 68\%$$

$$\mathbf{P}(|X - \mu| \leq 2\sigma) \approx 95\%$$

$$\mathbf{P}(|X - \mu| \leq 3\sigma) \approx 99.7\%$$

Business Interpretation:
Standard deviation directly quantifies risk and uncertainty.

Business Meaning of LLN

The Law of Large Numbers explains:

- Why large datasets reduce noise
- Why small samples are unreliable
- Why ML improves with more data

Large data stabilizes estimates. This comes from the following inequality which holds for all $\epsilon > 0$:

$$\mathbf{P}(|\bar{X}_n - \mu| > \epsilon) \leq \frac{V(\bar{X}_n)}{\epsilon^2}$$

Why Does the Average Look Normal?

Customer Spending Example
Individual spending is highly skewed.

Why Does the Average Look Normal?

Customer Spending Example
Individual spending is highly skewed.

However:

- The average of 500 customers
- Appears approximately normal

Why?

Central Limit Theorem

Let $X_1, \ldots, X_n$ be i.i.d. with

$$\mathbb{E}[X_i] = \mu, \quad \mathrm{Var}(X_i) = \sigma^2$$

Then:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \to \mathcal{N}(0, 1)$$

as $n \to \infty$.

Business Meaning of the CLT

For large n:

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Implications:

- Enables confidence intervals
- Enables A/B testing
- Uncertainty decreases with $\sqrt{n}$

Market Research Example

A survey of 1000 customers:

- 180 would buy the product

Estimated probability:

$$\hat{p} = 0.18$$

Is this the true purchase probability?

Population vs Sample

Population parameter:

$$p = \text{True purchase probability}$$

Sample estimate:

$$\hat{p} = \frac{\text{Number of buyers}}{n}$$

In practice, we only observe samples.

Confidence Interval

Using the CLT:

$$\bar{X} \pm 1.96\frac{\sigma}{\sqrt{n}}$$

Interpretation:
If we repeated the study many times, 95% of the intervals would contain the true parameter.

Exercise: Histogram with Relative Frequency

- Is the wage distribution symmetric, left-skewed, or right-skewed?
- Where are most wages concentrated?
- Are there extreme values or outliers?

Exercise: Histogram with Relative Frequency

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load data
df = pd.read_csv("Wage_merged.csv")

wages = df["wage"]

# Histogram with relative frequency
plt.hist(wages, bins=20, density=True)

plt.xlabel("Hourly Wage")
plt.ylabel("Relative Frequency")
plt.title("Distribution of Hourly Wages")

plt.show()
```

Exercise: Law of Large Numbers (LLN)

- What happens to the sample mean when the sample size increases?
- What happens to the variance of the estimator as the sample size grows?
- Are there extreme values or outliers?

Exercise: Law of Large Numbers (LLN)

```python
import numpy as np

wages = df["wage"]
true_mean = wages.mean()

sample_sizes = range(5, 500)
means = []

for n in sample_sizes:
    sample = wages.sample(n, replace=True)
    means.append(sample.mean())

plt.plot(sample_sizes, means, label="Sample Mean")
plt.axhline(true_mean, linestyle="--", label="True Mean")

plt.xlabel("Sample Size")
plt.ylabel("Mean Wage")
plt.title("Law of Large Numbers")

plt.legend()
plt.show()
```

Exercise: Central Limit Theorem (CLT)

- Is the sample means distribution symmetric, left-skewed, or right-skewed?
- What happens if we increase the sample size from 30 to 100?

Exercise: Central Limit Theorem (CLT)

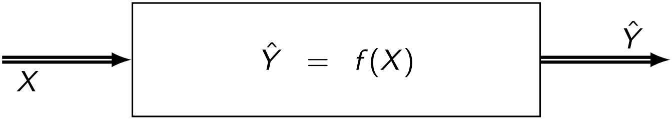
```python
sample_size = 30
num_samples = 1000

sample_means = []

for _ in range(num_samples):
    sample = wages.sample(sample_size, replace=True)
    sample_means.append(sample.mean())

plt.hist(sample_means, bins=30, density=True)

plt.xlabel("Sample Mean Wage")
plt.ylabel("Density")
plt.title("Central Limit Theorem")

plt.show()
```

Machine learning

Black box process

X, Y random variables.

Find a function f such that



and the probability of $(Y - \hat{Y})$ being large is not too big, i.e, expected prediction error (true loss)

$$\mathbf{E}[(Y - \hat{Y})^2] \text{ or more generally } \mathbf{E}[\ell(Y, \hat{Y})]$$

is small, $\ell(y, y')$ a measure of how close y and y' are.?

Machine learning

We do not fully know the distribution of X, Y.

$E[\ell(Y, \hat{Y})]$, $\hat{Y} = f(X)$ – depends on the distribution of X, Y

We know i.i.d. samples $x_1, \ldots, x_N$ of X and $y_1, \ldots, y_N$ of Y, x_i, y_i correspond to the same row (outcome)

Basic soluton: find f for which:

$$\frac{1}{N}\sum_{i=1}^{N} \ell(y_i, f(x_i)) + \text{some cleverly chosen term}$$

is small.

Question: will

$$E[(\ell(Y, f(X))]$$

be small, if $\frac{1}{N}\sum_{i=1}^{N} \ell(y_i, f(x_i)) + \text{some cleverly chosen term}$ is small ?

Tearser for future: generalisation bounds

If $x_1, x_2, \ldots, x_N$ i.i.d. sample of F, then with probability at least $1 - \delta$:

$$\underbrace{\mathbb{E}[\ell(Y, f(X))]}_{\text{true loss}} \leq \underbrace{\frac{1}{N}\sum_{i=1}^{N} \ell(y_i, f(x_i))}_{\text{empirical loss}} + \frac{C}{\sqrt{N}}\sqrt{\log(1/\delta)}$$

where C depends on the complexity of the model class $\mathcal{F}$, measured by:

- **VC dimension:** Growth rate of different models which behave the same way on finite samples
- **Rademacher complexity:** Average correlation with random labels

- Higher model complexity $\Rightarrow$ Larger C
- Larger sample size $N \Rightarrow$ Tighter bound (decreases as $1/\sqrt{N}$)
- The bound trades off: training error vs model complexity
- Smaller δ (higher confidence) $\Rightarrow$ Larger bound (looser guarantee)

Learning from Data: Estimating Distributions

Key Insight:
Estimating the mean, variance, and distribution from data is itself a form of machine learning.
Given i.i.d. samples: $x_1, x_2, \ldots, x_N$ from F.
We estimate the population parameters:
Empirical Mean:

$$\hat{m} = \frac{1}{N}\sum_{i=1}^{N} x_i \quad \text{estimates} \quad \mu = \mathbb{E}[X]$$

Empirical Distribution:

$$\hat{F}(a) = \frac{1}{N}\#\{i : x_i \leq a\} \quad \text{estimates} \quad F(a) = \mathbb{P}(X \leq a)$$

These are **learned models** from data, just like any other ML model.

Why hypothesis testing

Problem: Different datasets yield different model estimates.

Example:

- Dataset 1: Sample mean $\hat{m}_1 = 10.2$
- Dataset 2: Sample mean $\hat{m}_2 = 10.5$
- True population mean: unknown

Question: Is the difference real or just noise?

Hypothesis testing provides a framework to assess whether observed differences are statistically significant or due to random sampling variation.

Key Models We Estimate from Data:

- Empirical mean: $\hat{m} = \frac{1}{N} \sum x_i$
- Empirical variance: $\hat{\sigma}^2 = \frac{1}{N-1} \sum (x_i - \hat{m})^2$
- Empirical distribution: $\hat{F}(a) = \frac{\text{count}(x_i \leq a)}{N}$

Hypothesis Testing for the Mean

Setup: We observe i.i.d. samples $x_1, \ldots, x_N$ and want to test if the true mean is μ_0.

Null hypothesis: $H_0 : \mu = \mu_0$
Alternative hypothesis: $H_1 : \mu \neq \mu_0$

Test statistic:

$$t = \frac{\hat{m} - \mu_0}{\hat{\sigma}}$$

where $\hat{m} = \frac{1}{N} \sum x_i$ and $\hat{\sigma} = \sqrt{\frac{1}{N-1} \sum (x_i - \hat{m})^2}$.

Key fact (Central Limit Theorem):
Under H_0, the test statistic t approximately follows a Student's t-distribution (roughly normal for large N).

Decision Rule

We choose a significance level α (typically 0.05).

Critical value: Find c such that $\mathbb{P}(|t| > c) = \alpha$, if μ_0 is the true mean.

Rejection rule: Reject H_0 if $|t| > c$.

Interpretation:
If H_0 were true, observing $|t| > c$ would be very unlikely (probability α). Since we observed it, we reject H_0.

p-value: The smallest α for which we reject H_0. If p-value < 0.05, reject H_0.

Example: Mean Wage Test

Observed data: $\hat{m} = 10.5$, $\hat{\sigma} = 0.2$, $N = 100$

Test: Is the true mean wage equal to 10?

Test statistic:

$$t = \frac{10.5 - 10}{0.2} = 2.5$$

Critical value at $\alpha = 0.05$: $c \approx 1.98$

Decision: Since $|t| = 2.5 > 1.98$, we reject H_0.

Conclusion: The true mean wage is significantly different from 10 dollars.

Confidence Intervals

From the test statistic, we can construct a confidence interval.

Choose c such that $\mathbb{P}(|t| \leq c) = 1 - \alpha$.

Then with probability $1 - \alpha$:

$$\hat{m} - c \cdot \hat{\sigma} \leq \mu \leq \hat{m} + c \cdot \hat{\sigma}$$

Interpretation:
The confidence interval contains all values μ_0 for which we cannot reject $H_0 : \mu = \mu_0$ at significance level α.

Example: If the 95% confidence interval is $[10.1, 10.9]$, we can reject $H_0 : \mu = 10$ but not $H_0 : \mu = 10.5$.

Two-Sample Hypothesis Testing

Problem: Compare means of two independent groups.

Data:

- Group 1: $x_1, \ldots, x_N$ with mean $\hat{m}_1$ and variance $\hat{\sigma}_1^2$
- Group 2: $y_1, \ldots, y_M$ with mean $\hat{m}_2$ and variance $\hat{\sigma}_2^2$

Hypotheses:

$H_0 : \mu_1 = \mu_2$ (means are equal)
$H_1 : \mu_1 \neq \mu_2$ (means differ)

Two-Sample t-Test

Test statistic:

$$t = \frac{\hat{m}_1 - \hat{m}_2}{\sqrt{\frac{\hat{\sigma}_1^2}{N} + \frac{\hat{\sigma}_2^2}{M}}}$$

Under H_0, t approximately follows a t-distribution (by CLT).

Decision: Reject H_0 if $|t| > c$ (critical value at significance level α).

Business Example:
Compare average wages between men and women. If $|t|$ is large, evidence suggests systematic wage differences.

Example: Mean wage of 10 dollars

Observed data: $\hat{m} = 10.5$, $\hat{\sigma} = 0.2$, $N = 100$

Test statistic:

$$t = \frac{10.5 - 10}{0.2} = \frac{0.5}{0.2} = 2.5$$

Critical value at $\alpha = 0.05$: $c \approx 1.98$ (from t-distribution with 99 degrees of freedom)

Decision: Since $|t| = 2.5 > 1.98$, we reject H_0.

Conclusion: The evidence strongly suggests that the true mean wage is not 10 dollars; it is likely higher.

Examples

Example (t-test): Let $x_1, \dots, x_N$ be an i.i.d. sample with unknown mean μ and variance σ^2.
The null hypothesis is $H_0 : \mu = 0$.

$$\mathbf{T} = \hat{m} = \frac{1}{N} \sum_{i=1}^{N} x_i$$

$$\hat{\sigma} = \sqrt{\frac{1}{N(N-1)} \sum_{i=1}^{N} (x_i - \hat{m})^2}$$

The test statistic

$$\frac{\mathbf{T}}{\hat{\sigma}}$$

follows a Student's t-distribution. For large N, it is approximately normal.

Other examples include the Kolmogorov-Smirnov test, Bernoulli test, etc.

Python: scipy . stats .ttest_1samp, scipy . stats .ks_1samp,

Extensions of hypothesis testing

Non-zero null hypothesis: $H_0 : \theta = \theta_0$

One-sided alternative: Instead of $H_1 : \theta \neq \theta_0$, use $H_1 : \theta > \theta_0$ or $H_1 : \theta < \theta_0$.
We reject H_0 if $\mathbf{T} > c$ or $\mathbf{T} < c$ (respectively) instead of $|\mathbf{T}| > c$.

Two-sample hypothesis test:

- Let $x_1, \ldots, x_N$ be an i.i.d. sample with parameter θ_1.
- Let $y_1, \ldots, y_M$ be an i.i.d. sample with parameter θ_2.
- $H_0 : \theta_1 = \theta_2$, $H_1 : \theta_1 \neq \theta_2$.

We reject H_0 if $|\mathbf{T}| > c$, where the distribution of $\mathbf{T}$ under H_0 is known.
Examples: Kolmogorov-Smirnov test, t-test, Spearman's correlation, Pearson's correlation.
Python: `scipy.stats.ttest_2samp`, `scipy.stats.ks_2samp`, `scipy.stats.spearmanr`.

Confidence intervals

Choose c such that

$$\left| \frac{\mathbf{T} - \theta}{\hat{\sigma}} \right| > c$$

with probability α.
Then with probability $1 - \alpha$:

$$\mathbf{T} - c\hat{\sigma} \leq \theta \leq \mathbf{T} + c\hat{\sigma}$$

Confidence intervals

The probability of observing data such that θ falls outside the interval

$$[\mathbf{T} - c\hat{\sigma}, \mathbf{T} + c\hat{\sigma}]$$

is α.

This interval is the **confidence interval** at significance level α (or confidence level $1 - \alpha$).

Typically $\alpha = 0.02$ or $\alpha = 0.05$.

For any θ_* in the confidence interval, we cannot reject the null hypothesis $H_0 : \theta = \theta_*$ at significance level α.

If the confidence interval contains 0, we cannot reject that $\theta = 0$.

Exercise: Confidence Interval for the Mean Wage

- If we collect new wages different from the dataset, what would be the range of values with 95% of confidence ?

Exercise: Confidence Interval for the Mean Wage

```
wages = df["wage"]

n = len(wages)
mean_wage = wages.mean()
std_wage = wages.std(ddof=1)

# Standard error
se = std_wage / np.sqrt(n)

# 95% confidence interval
ci = stats.t.interval(0.95, df=n-1, loc=mean_wage, scale=se)

print("Mean wage:", mean_wage)
print("95% confidence interval:", ci)
```

Exercise: One-Sample Hypothesis Test

- If we collect new wages different from the dataset, what would be the range of values with 95% of confidence ?

Exercise: One-Sample Hypothesis Test

```
from scipy.stats import ttest_1samp

t_stat, p_value = ttest_1samp(wages, 10)

print("t-statistic:", t_stat)
print("p-value:", p_value)
```

Exercise: Two-Sample Hypothesis Test

- Are wages significantly different between men and women?
- What is the null hypothesis?
- What does the p-value tell us?

Solution: Two-Sample t-Test

```
from scipy import stats

men = df[df["female"] == 0]["wage"]
women = df[df["female"] == 1]["wage"]

# Perform two-sample t-test
t_stat, p_value = stats.ttest_ind(men, women)

print(f"Mean wage (men): {men.mean():.2f}")
print(f"Mean wage (women): {women.mean():.2f}")
print(f"t-statistic: {t_stat:.4f}")
print(f"p-value: {p_value:.4f}")

if p_value < 0.05:
    print("Reject H0: Wages are significantly different")
else:
    print("Fail to reject H0: No significant difference")
```

Exercise: Comparing Two Groups (Two-Sample Test)

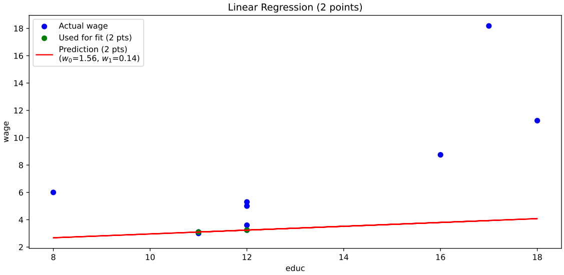
```python
men = df[df["female"] == 0]["wage"]
women = df[df["female"] == 1]["wage"]

from scipy.stats import ttest_ind

t_stat, p_value = ttest_ind(men, women)

print("t-statistic:", t_stat)
print("p-value:", p_value)
```

Module 2 — Linear Models

- Linear Regression (One Variable)
- Multiple Linear Regression
- Likelihood and Loss Functions

Linear Regression (1 Variable)

Model: $\hat{y} = f(x) = w_0 + w_1 x$



I.i.d sample $\{(x_i, y_i)\}_{i=1}^n$ from the distribution of Y, X.

Loss function (Least Squares): $\ell(y, y') = (y - y')^2$

Linear least squares (LLS): Find w_0, w_1 by minimizing

$$L(w_0, w_1) = \frac{1}{n}\sum_{i=1}^{n}(y_i - w_0 - w_1 x_i)^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \approx \mathbb{E}[(Y - f(X))^2]$$

Single Variable Linear Regression: intuition

Making

$$L(w_0, w_1) = \frac{1}{2}\sum_{i=1}^{2}(y_i - w_0 - w_1 x_i)^2$$

small is like solving equation

$$y_1 = w_0 + w_1 x_1, \quad y_2 = w_0 + w_1 x_2$$

for w_0 and w_1.

Example: Suppose we have two data points: $(x_1, y_1) = (1, 2)$ and $(x_2, y_2) = (3, 4)$. Solve to equations:

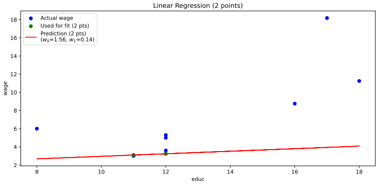
$$\begin{cases} 2 = w_0 + w_1 \cdot 1 \\ 4 = w_0 + w_1 \cdot 3 \end{cases}$$

$w_0 = w_1 = 1$

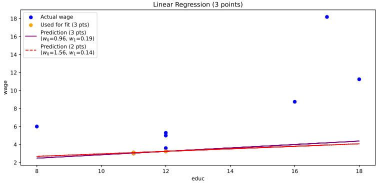
Result: The best-fit line is $\hat{y} = 1 + 1 \cdot x$.

Single Variable Linear Regression: intuition

For 2 data points: always a solution (line passes through both points)



For 3 data points: no solution.



Learning as minimization

Goal: minimize $L(w_0, w_1)$
Gradient descent: for simplicity $w_0 = 0$, i.e., $L(w_1) = L(\theta)$

$$\theta(t+1) = \theta(t) - \eta \nabla L(\theta(t))$$

η learning rate.

Stop when $\theta(t)$ does not change: $\min_\theta L(\theta) \approx L(\theta(t))$.

Gradient (derivative):

$$\nabla L(\theta) = \frac{L(\theta)}{d\theta} \approx \frac{L(\theta + \Delta) - L(\theta)}{\Delta}, \quad \Delta \text{ is very small}$$

Gradient points in the direction of steepest increase.

To minimize the loss, we move opposite to the gradient direction.

Example: Gradient for Linear Regression

For LSS: $\frac{L(\theta)}{d\theta} = \frac{2}{n} \sum_{i=1}^{n} (y_i - \theta x_i) x_i$

$$\theta(t+1) = \theta(t) + \frac{2\eta}{n} \sum_{i=1}^{n} \underbrace{(y_i - \theta(t) x_i)}_{\text{prediction error}} \; x_i$$

Choice of learning rate η:

- Too small $\rightarrow$ slow convergence
- Too large $\rightarrow$ divergence or oscillation
- Well chosen $\rightarrow$ fast and stable convergence

Learning rate is a key hyperparameter.

Variants of Gradient Descent

Classical gradient descent:

- Uses entire dataset at each iteration

In large datasets, we use:

- Stochastic Gradient Descent (SGD)
- Mini-batch Gradient Descent

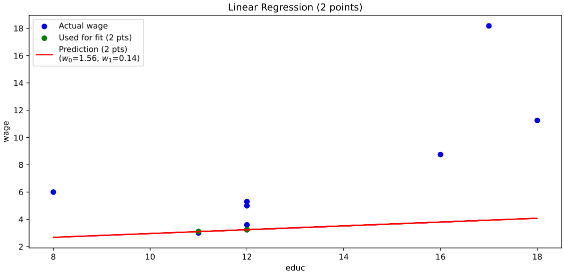
- **Batch size:** Number of samples processed before the model is updated.
- **Epoch:** One full iteration over all training data.

Example: If you have 1000 samples, batch size 100, then 1 epoch = 10 batches.

These methods are standard in deep learning.

Overfitting

Fits the *training data perfectly* but fails to generalize to new data:



Don't trust performance on training data: need test data.

Underfitting: does not even work on training data.

Avoid overfitting: large training sample, simple model (Occam's razor).

Multiple Linear Regression

Model:
$\hat{y} = f(x_1, \ldots, x_p) = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_p x_p$
Learning from $\{(x_{i1}, \ldots, x_{ip}, y_i)\}_{i=1}^n\}$

Loss function: $\ell(y, y') = (y - y')^2$

$$L(w_0, w_1, \ldots, w_p) = \frac{1}{n} \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - w_0 - \sum_{j=1}^{p} w_j x_{ij} \right)^2$$

Learning: minimize $L(w_0, w_1, \ldots, w_p)$

Gradient Descent for Several Variables

Gradient descent: $L(w_0, \ldots, w_p) = L(\theta)$

$$
\begin{aligned}
w_0(t+1) &= w_0(t) - \eta \frac{\partial L(w_0(t), \ldots, w_p(t))}{\partial w_0}, \\
w_1(t+1) &= w_1(t) - \eta \frac{\partial L(w_0(t), \ldots, w_p(t))}{\partial w_1}, \\
\ldots\ldots\ldots & \\
w_p(t+1) &= w_p(t) - \eta \frac{\partial L(w_0(t), \ldots, w_p(t))}{\partial w_p}
\end{aligned}
$$

η learning rate.

Stop when $w_0(t), \ldots, w_p(t)$ does not change:
$\min_\theta L(\theta) \approx L(w_0(t), \ldots, w_p(t))$.

Gradient points in the direction of steepest increase.

To minimize the loss, we move opposite to the gradient direction.

Gradient Descent for Several Variables

Multivariate Linear Regression

$$L(w_0, w_1, \ldots, w_p) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - w_0 - \sum_{j=1}^{p} w_j x_{ij}\right)^2$$

Gradient descend

$$w_0(t+1) = w_0(t) + \eta\frac{1}{n}\sum_{i=1}^{n}\left(y_i - w_0(t) - \sum_{j=1}^{p} w_j(t) x_{ij}\right),$$

$$w_1(t+1) = w_1(t) + \eta\frac{1}{n}\sum_{i=1}^{n}\left(y_i - w_0(t) - \sum_{j=1}^{p} w_j(t) x_{ij}\right) x_{i1},$$

$$\cdots\cdots\cdots$$

$$w_p(t+1) = w_p(t) + \eta\frac{1}{n}\sum_{i=1}^{n}\left(y_i - w_0(t) - \sum_{j=1}^{p} w_j(t) x_{ij}\right) x_{ip}$$

Statistical properties of linear least squares

What can we say about *true loss*

$$E[(Y - f(X))^2] = E[[Y - w_0 - \sum_{i=1}^{p} w_i X_i]^2]$$

average performance on unseen data.

Assume

$$Y = w_0 + w_{1,true} X_1 + \cdots + w_{p,true} X_p + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

i.e., there exists an 'ideal' linear regression.

$$E[(Y - f(X))^2] = E[\left((w_0 - w_{0,true}) + \sum_{i=1}^{p} (w_i - w_{i,true}) X_i \right)^2] + \sigma^2]$$

i.e. *true loss* is determined by *parameter estimation error*

$$w_0 - w_{0,true}, \quad w_1 - w_{1,true}, \quad \ldots, \quad w_p - w_{p,true}$$

and variance σ^2 of the error.

Statistical properties of linear least squares

Key observation:
The estimated coefficients $w_0, w_1, \ldots, w_p$ are computed from random data, so they are themselves random variables.

Different datasets $\rightarrow$ Different estimates.

Question:
What is the distribution of $w_i - w_{i,true}$ for $i = 0, 1, \ldots, p$?

Statistical properties of least squares estimation

- **Unbiasedness:**

$$E[w_i] = w_{i,true}, \quad i = 0, 1, \ldots, p$$
$$E[\underbrace{L(w_0, \ldots, w_p)}_{\text{empirical loss}}] = \sigma^2$$

- **Consistency:**
  As $n \to \infty$, $w_i \to w_{i,true}$ and $\hat{\sigma}^2 = \underbrace{L(w_0, \ldots, w_n)}_{\text{empirical loss}} \to \sigma^2$.

- **Known distribution:**
  The distribution of $w_i - w_{i,true}$ for $i = 0, 1, \ldots, p$ is known (Student's t-distribution).

Measuring goodness of fit: R^2

Let:

- $\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$ (sample mean)
- $\hat{y}_i = \hat{w}_0 + \sum_{j=1}^{p} \hat{w}_j x_{ij}$ (predicted value)

The R^2 (coefficient of determination):

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

Interpretation:

- $R^2 = 0$: Model explains no variance
- $R^2 = 1$: Perfect fit
- $0 < R^2 < 1$: Model explains fraction R^2 of variance

Hypothesis testing and confidence intervals for regression

Setup:
We test whether a coefficient equals zero:

$$H_0 : w_{j,true} = 0 \quad \text{vs} \quad H_1 : w_{j,true} \neq 0$$

Test statistic:

$$t = \frac{w_j - 0}{\mathrm{SE}(w_j)}$$

where $\mathrm{SE}(w_j)$ is the standard error (estimated standard deviation) of w_j.

Under H_0, t follows a Student's t-distribution.

Confidence interval for regression coefficients

Choose significance level α (typically 0.05).

The $(1-\alpha)$-confidence interval for $w_{j,true}$:

$$w_j \pm c \cdot \mathrm{SE}(w_j)$$

where c is the critical value from the t-distribution.

Interpretation:
All values $w^*_{j,true}$ in the confidence interval cannot be rejected at significance level α.

Practical

```
Python computes this automatically:
import statsmodels.api as sm

# Assuming X is the design matrix and y is the response
↪  variable
X = sm.add_constant(X)  # Adds a constant term to the predictor
model = sm.OLS(y, X).fit()  # Fit the model
print(model.summary())  # Print the summary of the regression
↪  results
```

Hypothesis Testing for MSE of Linear Regression

Two models w_1 and w_2

$L_1(w_1)$ on test set 1 and $L_2(w_2)$ on test set 2

Question: Are they the same ?

Approach:

1. For each model j, compute prediction errors $e_{i,j} = y_{i,j} - f_{w_j}(x_{i,j})$ on its test set.
2. Compute losses: $s_{i,j} = (y_{i,j}, f_{w_j}(x_{i,j}))^2$.
3. Compare the means $L_1(w_1) = \frac{1}{n_1}\sum_{i=1}^{n_1} s_{i,1}$ and $L_2(w_2) = \frac{1}{n_2}\sum_{i=1}^{n_2} s_{i,2}$ using a two-sample t-test:

$$t = \frac{L_1(w_1) - L_2(w_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

   where s_j^2 is the sample variance of $\{s_{i,j}\}$.
4. Compute the p-value; reject H_0 (no difference in expected loss) if $p < 0.05$.

Interpretation: If H_0 is rejected, the difference is significant.

Logistic Regression

Logistic regression is used for binary classification. The model predicts the probability that $y = 1$ given input x:

$$f(x) = \begin{cases} 1 & \text{if } \sigma(w_0 + \sum_i w_i x_i) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid (logistic) function.

Natural $0-1$ loss

$$L(w) = \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, f(x_i)),\ \ell(y, y') = \begin{cases} 0 & \text{if } y = y' \\ 1 & \text{if } y \neq y' \end{cases}$$

Cross-Entropy Loss Function

For n samples (x_i, y_i) with $y_i \in \{0, 1\}$, the cross-entropy loss is:

$$L(w) = -\frac{1}{n}\sum_{i=1}^{n} \left[y_i \log p_i + (1 - y_i)\log(1 - p_i)\right]$$

where $p_i = \sigma(w_0 + \sum_j w_j x_{ij})$.

Evaluation Metrics

- **True Positives (TP)**: Number of instances correctly predicted as positive.
- **False Positives (FP)**: Number of instances incorrectly predicted as positive (actually negative).
- **False Negatives (FN)**: Number of instances incorrectly predicted as negative (actually positive).

These metrics are commonly used in classification tasks to evaluate the performance of a model.

- **Accuracy**: Fraction of correct predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \approx 1 - E[\ell(Y, f(X))]$$

where $\ell(y, y') = \begin{cases} 0 & \text{if } y = y' \\ 1 & \text{if } y \neq y' \end{cases}$.

- **Recall**: Fraction of actual positives correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **Precision**: Fraction of predicted positives that are correct.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **F1-score**: Harmonic mean of precision and recall.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Python Code Example: Logistic Regression

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, recall_score,
↪  precision_score, f1_score

# X: feature matrix, y: binary labels
model = LogisticRegression()
model.fit(X, y)
y_pred = model.predict(X)

print("Accuracy:", accuracy_score(y, y_pred))
print("Recall:", recall_score(y, y_pred))
print("Precision:", precision_score(y, y_pred))
print("F1-score:", f1_score(y, y_pred))
```

Machine learning: general setup

Model class: $\hat{y} = f_w(x)$

Loss $\ell(y, \hat{y})$

Sample $\{(x_i, y_i)\}_{i=1}^n$ from the distribution Y and X

Learning objective: minimize the *empirical loss*

$$\min_w L(w)$$

where

$$L(w) = \frac{1}{n} \sum_{i=1}^{n} \ell(y_i, f_w(x_i))$$

Motivation: small $L(w)$ implies small *true loss*

$$E[\ell(Y, f_w(X))]$$

Idea of Gradient Descent

To solve:

$$\min_{w} L(w)$$

we use gradient descend

$$w_{t+1} = w_t - \eta \nabla L(w_t)$$

We subtract the gradient (scaled by learning rate η) to move downhill toward lower loss values.

- η = learning rate
- $\nabla L(w)$ = gradient (derivative)
- Repeat until convergence

Choice of learning rate η:

- Too small $\rightarrow$ slow convergence
- Too large $\rightarrow$ divergence or oscillation
- Well chosen $\rightarrow$ fast and stable convergence

Learning rate is a key hyperparameter.

Variants of Gradient Descent

Classical gradient descent:

- Uses entire dataset at each iteration

In large datasets, we use:

- Stochastic Gradient Descent (SGD)
- Mini-batch Gradient Descent

These methods are standard in deep learning.

Module 5 — Nonlinear Models

- Decision Trees
- Random Forest
- Introduction to Neural Networks

Decision Trees

A decision tree is a predictive model that splits data into branches based on feature values, forming a tree-like structure. Each internal node represents a decision based on a feature, each branch corresponds to an outcome of that decision, and each leaf node represents a final prediction (class or value). The tree recursively partitions the data to maximize homogeneity within leaves, making decisions interpretable and easy to visualize.

Example: Decision Tree for Loan Approval

Suppose a bank wants to automate loan approvals. The decision tree might split as follows:

- **Node 1:** Is credit score $>$ 700?
  - **Yes:** Approve loan
  - **No:** Go to Node 2
- **Node 2:** Is annual income $>$ \$50,000?
  - **Yes:** Approve loan
  - **No:** Reject loan

Credit score $>$ 700?

Random Forest: Main Idea

Instead of one tree:

Build many trees

Each tree

- Trained on bootstrap sample of data
- Uses random subset of features at each split

Aggregation

- Regression $\rightarrow$ average prediction
- Classification $\rightarrow$ majority vote

Why Random Forest Works

Each individual tree:

- Low bias
- High variance

Averaging many de-correlated trees:

Reduces variance

Ensemble of unstable models $\rightarrow$ Stable predictor

Random Forest and Bias–Variance

Compared to a single tree:

- Similar bias
- Much lower variance

Random Forest:

Improves generalization performance

Often works well with minimal tuning.

Motivation for Boosting

Suppose a small tree is weak:

- Slightly better than random
- High bias

Key question:

Can many weak learners create a strong model?

Boosting: Main Idea

Core mechanism

Train models sequentially

Each new tree:

- Focuses on previous errors
- Corrects residual mistakes

Final model:

$$F(x) = \sum_{m=1}^{M} \gamma_m T_m(x)$$

Boosting Intuition

Step 1:

- Fit a small tree

Step 2:

- Identify errors
- Increase focus on difficult observations

Step 3:

- Fit new tree to residual errors

Repeat until performance stabilizes.

Boosting as Gradient Descent

Boosting can be interpreted as:

Gradient descent in function space

At each iteration:

- Compute residuals (negative gradient)
- Fit tree to residuals
- Update model

Examples: Gradient Boosting, XGBoost, LightGBM

Random Forest vs Boosting

Random Forest

- Trees trained independently
- Mainly reduces variance
- Easily parallelizable

Boosting

- Trees trained sequentially
- Mainly reduces bias
- Often higher predictive performance

Introduction to Neural Networks

One hidden layer network:

$$h = \sigma(W_1 x + b_1)$$

$$\hat{y} = W_2 h + b_2$$

Key idea

- Linear transformations
- Followed by nonlinear activation

Universal function approximator when sufficiently large.

Module 6 — ML in Practice

- Feature Engineering
- Train / Validation / Test Split
- Cross-Validation
- Practical ML Pipeline
- Model Monitoring
- Hypothesis Testing for Model Drift

Business Example: Raw Data Is Not Enough

Suppose we predict house prices.
Available variables:

- Size (m^2)
- Year built
- Zip code

Question:
Should we use them directly?

Feature Engineering

Feature engineering:

Transform raw data into informative variables

Examples:

- Age of house = Current year - Year built
- One-hot encoding of zip code
- Interaction: Size $\times$ Location
- Log-transform price

Better features $\rightarrow$ Better models

Common Feature Transformations

- Scaling (standardization)

$$x' = \frac{x - \mu}{\sigma}$$

- Log-transform for skewed variables
- Polynomial features
- Categorical encoding (one-hot)

Important for:

- Linear models
- Regularization
- Gradient descent

Why Train/Test Split?

Suppose our model achieves:

$$R^2 = 0.95$$

on training data.

Question:
Will it perform equally well on new customers?

Train / Validation / Test Split

Dataset is divided into:

- Training set $\rightarrow$ Fit model
- Validation set $\rightarrow$ Tune hyperparameters
- Test set $\rightarrow$ Final unbiased evaluation

Important:

Test set must not influence model choice

Cross-Validation

Instead of one split:

Use multiple splits

K-fold cross-validation:

- Split data into K folds
- Train on K-1 folds
- Validate on remaining fold
- Repeat K times

Provides more stable performance estimate.

Business Problem: Model Drift

Suppose churn model was deployed 6 months ago.

Question:
Has customer behavior changed?

If distribution changes:

$$P_{\text{train}}(X) \neq P_{\text{production}}(X)$$

Model performance may degrade.

Monitoring with Hypothesis Testing

We test:

$$H_0 : \text{Distribution unchanged}$$

$$H_1 : \text{Distribution changed}$$

Example tests:

- t-test (mean change)
- Kolmogorov–Smirnov test
- Chi-square test (categorical)

Monitoring Model Performance

We monitor metrics over time:

- Accuracy
- Precision / Recall
- R^2

Hypothesis test:

$$H_0 : \text{Performance unchanged}$$

If rejected:

- Retrain model
- Update features
- Re-evaluate pipeline

Typical ML Pipeline

1. Data collection
2. Feature engineering
3. Train/Validation split
4. Model training
5. Hyperparameter tuning
6. Final evaluation on test set
7. Deployment

Machine learning is an iterative process.

Module 4 — Generalization

- Overfitting vs Underfitting
- Bias–Variance Tradeoff
- Regularization (L1 and L2)
- Decision Boundaries
- Performance Metrics:
  - R^2
  - Accuracy
  - Precision
  - Recall
  - F1-score

Overfitting vs Underfitting

Problem: How well does a model generalize to unseen data?

- **Underfitting:** Model is too simple
  - Fails to capture patterns in training data
  - High bias, low variance
  - Poor performance on both training and test sets
  - Example: Linear model trying to fit a quadratic trend
- **Overfitting:** Model is too complex
  - Captures noise as if it were signal
  - Low bias, high variance
  - Excellent performance on training set, poor on test set
  - Example: High-degree polynomial for limited data points

Business Example: A pricing model fits historical sales perfectly (overfit) but predicts future sales poorly.

Bias–Variance Tradeoff

Problem: Why does increasing model complexity sometimes hurt performance?

Prediction error can be decomposed into:

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

- **Bias:** Error from overly simple assumptions High bias $\rightarrow$ underfitting
- **Variance:** Sensitivity to small fluctuations in training data High variance $\rightarrow$ overfitting
- **Noise:** Irreducible randomness in the data

Business Example: A very simple demand model ignores seasonality (high bias). A very complex model reacts to random weekly fluctuations (high variance).

Regularization

Goal: Prevent overfitting by controlling model complexity.

General form of regularized loss:

$$L(\mathbf{w}) = \sum_{i=1}^{n}(y_i - \mathbf{w}^T x_i)^2 + \lambda\Omega(\mathbf{w})$$

- λ controls the strength of the penalty
- Larger $\lambda \rightarrow$ simpler model

Method	Penalty	Effect
Ridge (L2)	$\sum w_j^2$	Shrinks coefficients
Lasso (L1)	$\sum \lvert w_j \rvert$	Produces sparse model

Business Example: In marketing analytics with 100 features, regularization avoids unstable and extreme coefficient estimates.

Decision Boundary in Logistic Regression

Goal: Understand how logistic regression separates classes.

Prediction rule:

$$Y = 1 \quad \text{if} \quad \sigma(w^T x) > 0.5$$

This is equivalent to:

$$w^T x = 0$$

- The equation defines a **hyperplane**
- Points on one side $\rightarrow$ Class 1
- Points on the other side $\rightarrow$ Class 0

Business Example: In credit scoring, the decision boundary separates high-risk from low-risk applicants based on their features.

Regression Metrics

- **R^2 (Coefficient of Determination):**

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

  Indicates proportion of variance explained by the model.
  $R^2 \approx 1$ is good.

- **Mean Squared Error (MSE):**

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

  Penalizes large errors more heavily.

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

  Robust to outliers; measures average absolute deviation.

Classification Metrics - Part 1

- **Accuracy:** Fraction of correct predictions

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

  Simple overall measure, but can be misleading for imbalanced classes.

- **Precision:** Fraction of positive predictions that are correct

$$\text{Precision} = \frac{TP}{TP + FP}$$

  Important when false positives are costly.

Classification Metrics - Part 2

- **Recall (Sensitivity):** Fraction of actual positives correctly identified

$$\text{Recall} = \frac{TP}{TP + FN}$$

Important when missing a positive is costly.

- **F1-score:** Harmonic mean of Precision and Recall

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Balances precision and recall.

Module 7 — Clustering

- Introduction to Clustering
- K-Means Algorithm
- Importance of Distance Metrics
- Silhouette Score

Business Example: Customer Segmentation

Suppose we have customer data:

- Annual income
- Spending score

Goal:

Group customers into segments

No labels are available.

What is Clustering?

Clustering is an unsupervised learning task.

Given data points:

$$x_1, \ldots, x_n$$

Partition them into K groups such that:

- Points in same cluster are similar
- Points in different clusters are dissimilar

K-Means Algorithm

Choose number of clusters K.

Goal:

$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

Where:

$$\mu_k = \text{mean of cluster } C_k$$

K-Means: Algorithm Steps

1. Initialize K centroids randomly
2. Assign each point to closest centroid
3. Update centroids (compute cluster means)
4. Repeat until convergence

Convergence:

- Assignments stop changing
- Centroids stabilize

Importance of Distance

K-means relies on distance:

$$\|x_i - \mu_k\|$$

Typically:

Euclidean distance

Implications:

- Features must be scaled
- Sensitive to outliers
- Assumes spherical clusters

Why Feature Scaling is Crucial

Suppose we cluster customers using:

- Income (0–100,000)
- Spending score (0–100)

Without scaling:

$$\text{Income dominates distance}$$

Solution:

$$x' = \frac{x - \mu}{\sigma}$$

Choosing the Number of Clusters

Key question:

What is the optimal K?

Common methods:

- Elbow method
- Silhouette score

Silhouette Score

For each point:

- a = average distance to points in same cluster
- b = smallest average distance to other clusters

Silhouette value:

$$s = \frac{b - a}{\max(a, b)}$$

Interpretation of Silhouette Score

$$-1 \leq s \leq 1$$

- $s \approx 1 \rightarrow$ well clustered
- $s \approx 0 \rightarrow$ overlapping clusters
- $s < 0 \rightarrow$ likely misclassified

Overall quality:

Average silhouette score

Business Interpretation

High silhouette score:

- Clear customer segments
- Targeted marketing possible

Low silhouette score:

- Segments overlap
- Clustering may not be meaningful

Clustering must be both:

Mathematically sound and business relevant

Conclusion

- From Probability to Business Decisions
- Choosing the Right Model
- The Complete ML Pipeline
- Responsible and Monitored Deployment